

Unit 3 Guided Notes — Equations & Inequalities

Keep it balanced. Undo step by step. Flip the sign when you divide by a negative.

Standards: NC.8.EE.7 · NC.8.EE.7a · NC.8.EE.7b

This packet has three parts: **Part 1 — Guided Notes**, **Part 2 — Practice Problems**, and **Part 3 — Answer Key**. Fill the blanks on your printed copy first, then click any blank, step, or check button to reveal the answer.

Topics Covered: 3.1 One-Step Equations / Inverse Operations · 3.2 Two-Step Equations · 3.3 Distributive Property · 3.4 Variables on Both Sides · 3.5 Multi-Step Equations · 3.6 Special Cases (No Solution / Infinite Solutions) · 3.7 Linear Inequalities · 3.8 Word Problems (all NC.8.EE.7)

1 One-Step Equations / Inverse Operations

★ **INVERSE OPERATIONS** are used to isolate the variable. Whatever you do to one side, you **MUST** do to the other to keep the equation balanced!

INVERSE (UNDO) OPERATIONS



Use the inverse to 'undo' and isolate the variable.

Addition	Subtraction	Multiplication	Division
Inverse: Subtraction	Inverse: Addition	Inverse: Division	Inverse: Multiplication
$x + 5 = 12$	$x - 3 = 8$	$3x = 15$	$x/4 = 2$

Match the Operation to its Inverse

MATCH Click an operation, then its inverse.

Operation

Addition

Subtraction

Multiplication

Division

Inverse

Division

Subtraction

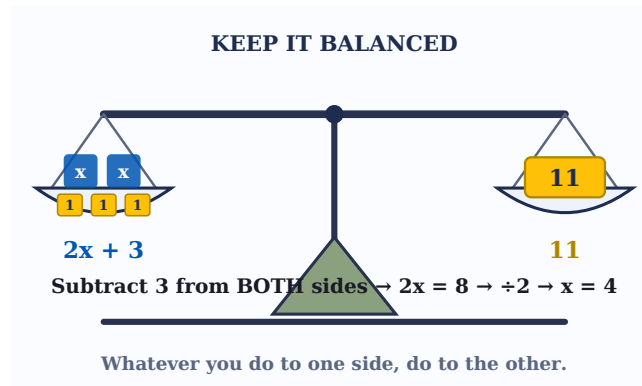
Multiplication

Addition

0 of 4 matched

2 Two-Step Equations

✦ **TWO STEPS:** Step 1 — Undo addition/subtraction | Step 2 — Undo multiplication/division



Example 1: $-2x + 4 = 8$

Step 1 – Undo addition (subtract 4): $-2x + 4 - 4 = 8 - 4 \rightarrow -2x = 4$

✓ Step 2 – Undo multiplication ($\div -2$): $x = -2$

Example 2: $w/2 + 6 = 7$

Step 1 – Undo addition (subtract 6): $w/2 = 1$

✓ Step 2 – Undo division ($\times 2$): $w = 2$

3 Distributive Property

✦ **THE DISTRIBUTIVE PROPERTY:** Any number (or variable) outside of the parentheses is multiplied by everything inside the parentheses. Pay special attention to positive/negative signs!

Example: $2(2x + 3) = -6$

1. **Distribute:** $2(2x) + 2(3) = 4x + 6 = -6$

2. **Undo addition (subtract 6):** $4x = -12$

✓ 3. **Undo multiplication ($\div 4$):** $x = -3$

ERROR ANALYSIS A student expands $-3(x - 4)$ as $-3x - 12$. What went wrong?

A negative times a negative is positive: $-3 \cdot (-4) = +12$. Correct: $-3x + 12$. Distribute the sign to every term inside.

4 Variables on Both Sides

✦ **STRATEGY:** Move variables from the RIGHT to the LEFT using inverse operations, then solve like a two-step equation.

Example: $5x + 3 = 2x + 12$

Move variables (subtract $2x$): $3x + 3 = 12$

Undo addition (subtract 3): $3x = 9$

✓ **Undo multiplication ($\div 3$):** $x = 3$

5 Multi-Step Equations — The 5-Step Process

The 5 steps: 1. Distribute → 2. Combine Like Terms → 3. Move Variables → 4. Inverse for + or − → 5. Inverse for × or ÷

Example: $-2x + 1 = 8(x + 2) - 5x$

1. Distribute ($8 \cdot x$ and $8 \cdot 2$): $-2x + 1 = 8x + 16 - 5x$

2. Combine like terms ($8x - 5x$): $-2x + 1 = 3x + 16$

3. Move variables ($+2x$): $1 = 5x + 16$

4. Undo addition (-16): $-15 = 5x$

✓ 5. Undo multiplication ($\div 5$): $x = -3$

6 Special Cases — No Solution & Infinite Solutions

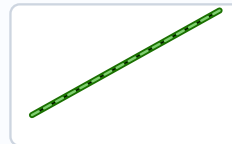
✦ **NO SOLUTION:** Variables cancel and you get a FALSE statement (like $5 = 9$).

✦ **INFINITE SOLUTIONS:** Variables cancel and you get a TRUE statement (like $6 = 6$).

WHEN THE VARIABLE DISAPPEARS



$6 = 10 \times$
FALSE → No Solution
lines never cross



$8 = 8 \checkmark$
TRUE → Infinite Solutions
same line (all points work)

One Solution	No Solution	Infinite Solutions
Variables stay; you get $x =$ number	Variables cancel; FALSE statement remains	Variables cancel; TRUE statement remains
$2x + 1 = 7 \rightarrow x = 3$	$2x + 1 = 2x + 5 \rightarrow 1 = 5 \times$	$2x + 1 = 2x + 1 \rightarrow 1 = 1 \checkmark$

No Solution — Solve: $3(x + 2) = 3x + 10$

Distribute & subtract $3x$: $3x + 6 = 3x + 10 \rightarrow 6 = 10 \rightarrow \text{FALSE! No Solution}$

Infinite Solutions — Solve: $2(x + 4) = 2x + 8$

Distribute & subtract $2x$: $2x + 8 = 2x + 8 \rightarrow 8 = 8 \rightarrow \text{TRUE! Infinite Solutions}$

Quick Check

MC Solving an equation gives $7 = 7$. The equation has...

✓ Variables cancelled and a TRUE statement remains → infinitely many solutions.

7 Linear Inequalities

Fill the meaning, circle, and shade on your printed copy, then click to check.

Symbol	Meaning	Circle	Shade
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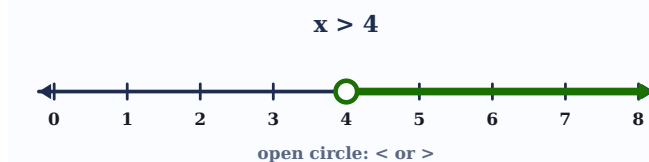
$<$			
$>$			
$\leq$			
$\geq$			

⚠ **CRITICAL RULE — FLIP THE SIGN:** When you MULTIPLY or DIVIDE both sides by a NEGATIVE number, you must FLIP the inequality sign! Example: $-2x > 6 \rightarrow$ divide by $-2 \rightarrow x < -3$ (sign flipped!)

Example (no flip): $2x + 3 > 11$

Subtract 3: $2x > 8$

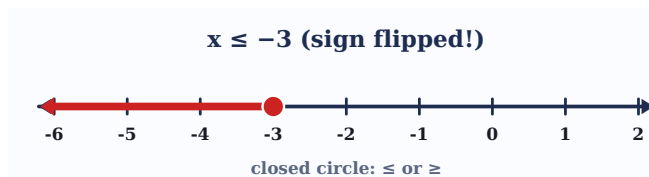
✓ Divide by 2 (positive – no flip): $x > 4$



Example (FLIP NEEDED): $-3x + 4 \geq 13$

Subtract 4: $-3x \geq 9$

✓ Divide by $-3 \rightarrow$ FLIP THE SIGN: $x \leq -3$



ERROR ANALYSIS To solve $-4x > 12$ a student wrote $x > -3$. What's the mistake?

Dividing both sides by the negative number -4 requires FLIPPING the sign. Correct: $x < -3$.

8 Word Problems — Equations & Inequalities

★ **WORD PROBLEM FLOWCHART:** 1) READ → 2) DEFINE the variable → 3) IDENTIFY keywords → 4) WRITE the equation/inequality → 5) SOLVE → 6) CHECK → 7) ANSWER in a sentence.

Equation (=)	Inequality (< > ≤ ≥)	Addition (+)	Subtraction (−)
"total," "equals," "is," "the same as," "gives," "results in"	"at least" (≥), "at most" (≤), "more than" (>), "fewer than" (<), "no more than" (≤)	sum, plus, more than, increased by, total	difference, minus, less than, decreased by

SHORT ANSWER **Problem 1 (Equation) — Movie Theater:** A movie theater charges a \$6 entry fee plus \$4 per ticket. If the total cost was \$30, how many tickets were purchased?

Let x = number of tickets → $6 + 4x = 30$. Solve: $4x = 24$ → $x = 6$ → **6 tickets** were purchased.

SHORT ANSWER **Problem 2 (Inequality) — Notebooks:** A student wants to spend no more than \$50 on notebooks that cost \$6 each. How many notebooks can the student buy?

Let x = number of notebooks → $6x \leq 50$ ("no more than" = $\leq$). Solve: $x \leq 8.33$ → **at most 8 notebooks**.

PART 2 — Practice Problems

Show all steps on your printed copy. Circle your final answer, then check on screen.

A Multi-Step Equations

Steps: 1) Distribute 2) Combine like terms 3) Move variables 4) Isolate the variable. (Type just the value, e.g. $x=5$ or 5.)

1 $3x + 7 = 22$

$x = ?$

2 $2(x - 4) = 10$

$x = ?$

3 $5x - 3 = 2x + 9$

$x = ?$

4 $-4(x + 2) = 12$

$x = ?$

5 $6x + 4 = 4x - 8$

$x = ?$

6 $3(2x + 1) = 4x + 7$

$x = ?$

7 $x/4 + 5 = 8$

$x = ?$

8 $7 - 2x = 3x + 17$

$x = ?$

9 $0.5x + 3 = 7$

$x = ?$

10 $-(x - 6) = 2x + 3$

$x = ?$

B Special Cases

Key: Variables cancel + TRUE → Infinite Solutions. Variables cancel + FALSE → No Solution.

1 $2x + 3 = 2x + 3$

✓ Both sides identical → all x work → Infinite Solutions.

2 $4x + 1 = 4x - 5$

✓ 4x cancels → $1 = -5$ FALSE → No Solution.

3 $3(x + 2) = 3x + 6$

✓ $3x + 6 = 3x + 6$ → always true → Infinite Solutions.

4 $5x - 5 = 5(x + 1)$

✓ $5x - 5 = 5x + 5$ → $-5 = 5$ FALSE → No Solution.

5 $7x + 2 = 7x + 2$

✓ Same expression both sides → Infinite Solutions.

6 $x + 4 = x - 1$

✓ x cancels → $4 = -1$ FALSE → No Solution.

C Inequalities — Solve

CRITICAL RULE: When you multiply or divide by a NEGATIVE number, FLIP the inequality sign!
(Type like $x > 3$ or $x \leq 6$.)

1 $3x + 2 > 11$

x ? value

2 $2x - 5 \leq 7$

x ? value

3 $-4x > 12$

x ? value

4 $5x + 1 \geq 16$

x ? value

5 $7 - x < 4$

x ? value

6 $2(x + 3) \leq 10$

x ? value

7 $-3x + 9 \geq 0$

x ? value

8 $4x - 1 > 2x + 5$

x ? value

D Word Problems

Strategy: Define a variable → Write an equation or inequality → Solve → Check if your answer makes sense!

1. A phone plan costs \$35/month plus \$0.10 per text. Last month's bill was \$52. How many texts were sent?

Let t = texts $\rightarrow 35 + 0.10t = 52$. Solve: $0.10t = 17 \rightarrow t = \mathbf{170 \text{ texts}}$.

2. Sam has \$240 and saves \$30/week. Jake has \$420 and spends \$10/week. When will they have the same amount?

Let w = weeks $\rightarrow 240 + 30w = 420 - 10w$. Solve: $40w = 180 \rightarrow w = \mathbf{4.5 \text{ weeks}}$.

3. A rectangle's perimeter is 56 cm. The length is 4 cm more than the width. Find the width.

Let w = width $\rightarrow 2(w + 4) + 2w = 56$. Solve: $4w + 8 = 56 \rightarrow w = \mathbf{12 \text{ cm}}$.

4. Movie tickets cost \$9 each. You have at most \$50 to spend. What is the maximum number of tickets?

Let t = tickets $\rightarrow 9t \leq 50$ ("at most" = $\leq$). Solve: $t \leq 5.56 \rightarrow \mathbf{\text{max 5 tickets}}$ (must be a whole number).

5. A car rental costs \$45/day plus \$0.25/mile. Your budget is \$120 for one day. How many miles can you drive?

Let m = miles $\rightarrow 45 + 0.25m \leq 120$. Solve: $0.25m \leq 75 \rightarrow m \leq \mathbf{300 \text{ miles}}$.

6. Three consecutive even integers sum to 78. Find the smallest integer.

Let n = smallest $\rightarrow n + (n+2) + (n+4) = 78$. Solve: $3n + 6 = 78 \rightarrow n = \mathbf{24}$.

★ PART 3 — Answer Key ★

Do not distribute to students until after the lesson.

★ Answers

Guided Notes

3.2: $-2x = 4 \rightarrow x = -2$; $w/2 = 1 \rightarrow w = 2$. 3.3: $4x + 6 = -6 \rightarrow 4x = -12 \rightarrow x = -3$. 3.4: $3x + 3 = 12 \rightarrow 3x = 9 \rightarrow x = 3$. 3.5: $8x + 16 - 5x \rightarrow 3x + 16 \rightarrow 5x + 16 \rightarrow -15 = 5x \rightarrow x = -3$. 3.6: No Solution ($6 = 10$ FALSE); Infinite Solutions ($8 = 8$ TRUE). 3.7: $<$ less than / OPEN / LEFT; $>$ greater than / OPEN / RIGHT; $\leq$ less than or equal / CLOSED / LEFT; $\geq$ greater than or equal / CLOSED / RIGHT. No-flip: $2x > 8 \rightarrow x > 4$. Flip: $-3x \geq 9 \rightarrow x \leq -3$.

Part A — Multi-Step

1) $x=5$ 2) $x=9$ 3) $x=4$ 4) $x=-5$ 5) $x=-6$ 6) $x=2$ 7) $x=12$ 8) $x=-2$ 9) $x=8$ 10) $x=1$.

Part B — Special Cases

1) Infinite Solutions 2) No Solution 3) Infinite Solutions 4) No Solution 5) Infinite Solutions 6) No Solution.

Part C — Inequalities

1) $x > 3$ 2) $x \leq 6$ 3) $x < -3$ (FLIP) 4) $x \geq 3$ 5) $x > 3$ (FLIP) 6) $x \leq 2$ 7) $x \leq 3$ (FLIP) 8) $x > 3$.

Part D — Word Problems

1) $35 + 0.10t = 52 \rightarrow t = 170$ texts 2) $240 + 30w = 420 - 10w \rightarrow w = 4.5$ weeks 3) $2(w+4) + 2w = 56 \rightarrow w = 12$ cm 4) $9t \leq 50 \rightarrow \text{max } 5$ tickets 5) $45 + 0.25m \leq 120 \rightarrow m \leq 300$ miles 6) $n + (n+2) + (n+4) = 78 \rightarrow n = 24$.

← Back to Unit 3